

# CHORUS: Checkpoint Orchestration with Rate-limiting and Unified Service-policies

Advanced Computing Project 25/26

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June 02, 2026

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# Context & Problem

- **Context:** LLM training generates massive checkpoints (GBs to TBs).
- **The Bottleneck:** Parallel File Systems (PFS) like Lustre suffer from synchronized I/O bursts ("Noisy Neighbor" problem).
- **Consequence:** Compute nodes stall while waiting for slow synchronous writes, wasting expensive resources.

## Central Insight

Checkpointing is **latency-critical for training continuation**, but not for long-term persistence.

# System Components

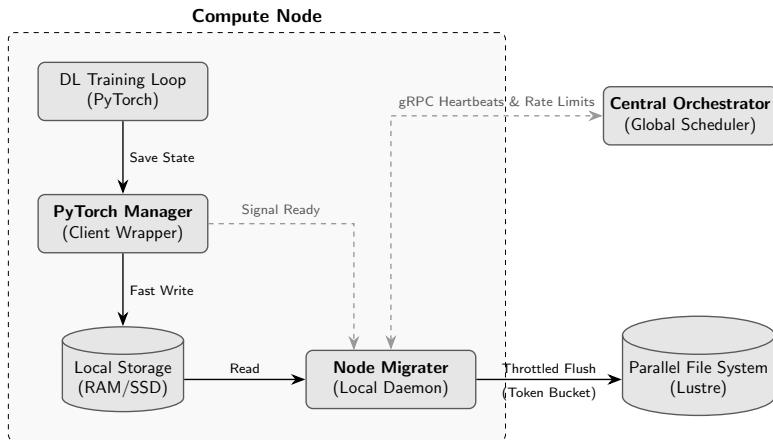


Figure 1. System architecture of the framework.

# Two-Tier Storage Strategy

## Local SSD Staging:

- Acts as a "Shock Absorber".
- Dedicated, uncontested bandwidth per node.
- Decouples training loop from network/PFS latency.

### Predictability Result

Writing to local SSDs showed **near-zero variance** in stall times compared to the high volatility of direct PFS writes.

# Pluggable Scheduling Policies

| Policy                    | Logic                                                                                      |
|---------------------------|--------------------------------------------------------------------------------------------|
| <b>Uniform Fair Share</b> | Splits bandwidth $B_{\text{PFS}}$ equally among all $N$ nodes, regardless of pending data. |
| <b>Active Fair Share</b>  | Splits bandwidth $B_{\text{PFS}}$ equally among nodes with pending data only.              |
| <b>Age Priority</b>       | Normalized combination of <i>pending size</i> and <i>waiting time</i> (anti-starvation).   |
| <b>Epoch Priority</b>     | Prioritizes jobs closer to completion (higher training progress $\rho_i$ ).                |

**Dynamic Control:** Orchestrator sends `CHANGE_RATE` or `HOLD` signals via persistent gRPC streams.

# Token Bucket & Traffic Shaping

- **Mechanism:** Migrater read/write operations consume tokens refilled at the Orchestrator-assigned rate.
- **Chunk Size Tuning:** Evaluated trade-off between throughput accuracy and PFS syscall pressure.

## Optimal Configuration

**4 MiB chunks** provided the best balance: high throughput accuracy with significantly reduced write frequency.

# Results: Throughput Profiles

## Burst Traffic

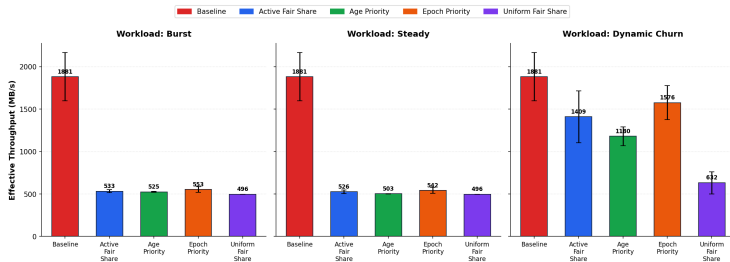
- Simultaneous dumps.

## Steady Traffic

- Staggered intervals.

## Dynamic Churn

- Job arrivals/departures.



## Setup Parameters:

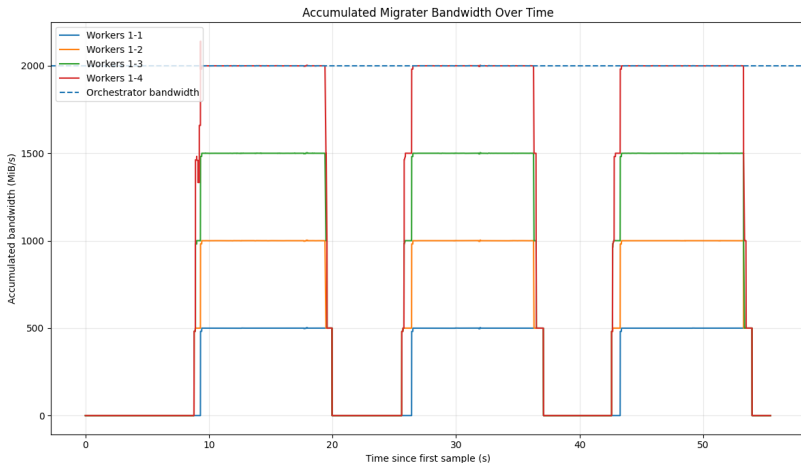
- 5 GB checkpoint payload.
- 2 GB/s PFS bandwidth.

## Cluster Environment:

- Tracked concurrent SLURM jobs to account for shared network background noise.

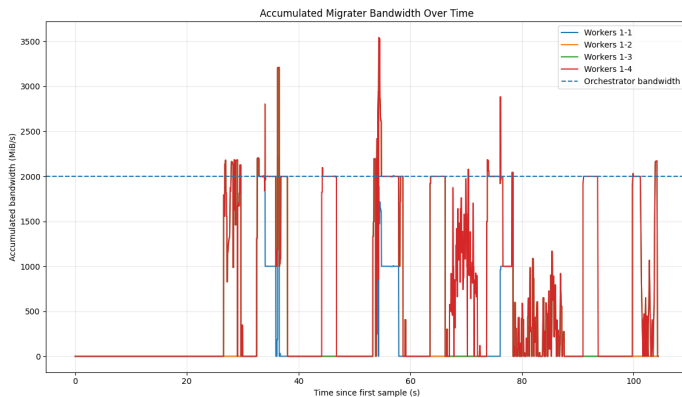


# Bandwidth Convergence



*The system successfully transforms uncoordinated I/O bursts into smooth, regulated storage traffic. However...*

# Bandwidth Convergence (Issues)



- **Signaling Latency Layer:** Caused by the 0.1s gRPC heartbeat sampling stream.
- **Token Accumulation Spikes:** Empty initial bucket balance prevents initial bursts, but sub-second spikes remain at block boundaries.

# Conclusions & Future Work

## Conclusions:

- Two-tier orchestration provides **predictability** for LLM training.
- Centralized rate control ensures **Cluster Harmony**.
- Local SSD staging is a scalable option to absorb bursty I/O patterns.

## Future Work:

- **Exascale:** Stress testing with thousands of concurrent nodes and weak scalability constraints.
- **Non-Blocking Queue Management:** Support queuing or pipelining new checkpoint requests if a worker generates a new state while the previous one is still migrating to the PFS.
- **Resilience:** Fault-tolerant Orchestrator (single point of failure).

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